

RAID (mdadm) & SMART Cheat Sheet

Check Linux software RAID, replace a failed disk, and spot dying drives before they fail.

RAID levels

Level	Min disks	Survives	Usable space	Good for
RAID 0	2	Nothing : one disk dies, all data is gone	All of it	Scratch space only
RAID 1	2	All but one disk	One disk	Boot / OS disks, small servers
RAID 5	3	1 disk	All but one disk	Read-heavy storage (slow rebuilds)
RAID 6	4	Any 2 disks	All but two disks	Big arrays of large disks
RAID 10	4	1 per mirror pair	Half	Databases, VMs (fast)

Reading cat /proc/mdstat and mdadm --detail

```
md0 : active raid1 sdb1[0] sdc1[1]
      976630464 blocks super 1.2 [2/2] [UU] ← 2 of 2 disks up: healthy
md1 : active raid5 sdg1[4] sdd1[0] sde1[1] sdf1[3](F) ← (F) = failed member · sdg1 = new disk
      1953260544 blocks super 1.2 level 5, 512k chunk, algorithm 2 [3/2] [UU_] ← _ = missing: degraded
      [==>.....] recovery = 17.3% (169005440/976630272) finish=72.4min speed=185632K/sec
$ sudo mdadm --detail /dev/md1 ← (excerpt) · the array works while rebuilding, but isn't safe until 100%
      State : clean, degraded, recovering ← clean = no half-done writes · degraded = a disk is missing
Active Devices : 2 ← in sync now · also listed: Working Devices : 3, Failed Devices : 1
  Number   Major   Minor   RaidDevice State
     0         8       49         0    active sync   /dev/sdd1 ← healthy member
     4         8       97         2    spare rebuilding /dev/sdg1 ← new disk being filled
     3         8       81         -    faulty        /dev/sdf1 ← the disk to replace
```

Replacing a failed disk (md0, bad disk sdc, good disk sdb)

Step	Command
1. Mark it failed and take it out	sudo mdadm /dev/md0 --fail /dev/sdc1 --remove /dev/sdc1
2. Note its serial, then swap the disk	sudo smartctl -i /dev/sdc grep Serial
3. Copy the partition layout from the good disk	sudo sfdisk -d /dev/sdb sudo sfdisk /dev/sdc
4. Add the new partition: rebuild starts	sudo mdadm /dev/md0 --add /dev/sdc1
5. Watch the rebuild	watch cat /proc/mdstat

GPT: after step 3 run **sgdisk -G /dev/sdc** for new unique IDs. Boot disk? Also **grub-install /dev/sdc**.

SMART: drive health

Command	Does	Command	Does
smartctl -H /dev/sda	PASSED / FAILED	smartctl -t short /dev/sda	~2 min test (long : hours)
smartctl -a /dev/sda	Everything	smartctl -l selftest /dev/sda	Self-test results

ID#	ATTRIBUTE_NAME	VALUE	WORST	THRESH	RAW_VALUE	← smartctl -a (columns cut) · RAW_VALUE = the real count
5	Reallocated_Sector_Ct	100	100	010	8	← bad sectors swapped for spares: rising = replace, even if -H says PASSED
187	Reported_Uncorrect	100	100	000	0	← errors the drive couldn't fix
197	Current_Pending_Sector	100	100	000	2	← unreadable sectors waiting: bad sign
198	Offline_Uncorrectable	100	100	000	2	← bad sectors found by a self-test
199	UDMA_CRC_Error_Count	200	200	000	143	← cable or connector errors, usually not the disk

Available Spare:	100%	← NVMe (smartctl -a /dev/nvme0): spare blocks left
Percentage Used:	3%	← wear; 100% = rated life used up
Media and Data Integrity Errors:	0	← should stay 0

Handy recipes

Command	Does
sudo mdadm --create /dev/md0 -l 1 -n 2 /dev/sdb1 /dev/sdc1	Makes a RAID 1 from 2 partitions
sudo mdadm --detail --scan sudo tee -a /etc/mdadm/mdadm.conf	Saves it (RHEL: /etc/mdadm.conf)
sudo update-initramfs -u	Assembles it at boot (RHEL: dracut -f)
sudo systemctl enable --now smartd	SMART alerts (Debian: auto-on)

Remember: RAID is not a backup. It survives a dead disk, not **rm -rf**, ransomware or a fire.